

Vishwash Prasad Joshi

+91 6266411668 contact@vpjoshi.in linkedin.com/in/vishwash-joshi vpjoshi.in

EDUCATION

B.Tech in Computer Science (AI/ML) - CGPA: 7.6/10 Aug 2023 – Jun 2027
RGPV Bhopal, Madhya Pradesh, India
Relevant Coursework: Operating Systems, DBMS, Computer Networks, OOPS, Data Structures and Algorithms

Class XII (CBSE) - 81% Jul 2022 – Jun 2023
Army Public School No. 2, Jabalpur, Madhya Pradesh, India
Highlights: Topped Mathematics, Physics, and Computer Science

TECHNICAL SKILLS

Languages: C++, Java, Python, Go, TypeScript, Bash
Cloud & DevOps: AWS, GCP, Docker, Kubernetes, Terraform, Linux, GitHub Actions
Backend & Frameworks: Node.js, FastAPI, React, Kafka, REST APIs, Microservices, CI/CD
Databases: PostgreSQL, MongoDB, DynamoDB
Authentication & APIs: OAuth 2.0, JWT, OpenAI API, Gemini API

WORK EXPERIENCE

Hartalkar Innovation Pvt Ltd, Jabalpur – Software Developer Intern Jun 2024 – Sept 2024

- Compiled a custom Linux kernel for EV hardware, reducing boot time by 90% by stripping non-essential modules.
- Designed a real-time vehicle monitoring UI in React.js with WebSocket-based live diagnostics.
- Developed a hardware-to-server telemetry pipeline (Arduino → Node.js) using the UART protocol.
- Developed a 3D collision avoidance simulation in Unity, integrating live vehicle sensor data to test auto-stop logic.

PROJECTS

Distributed Container Orchestrator — *Go, Docker, RESTful API*  [GitHub](#),  [Docs](#)

- Built a distributed container orchestrator to manage multi-node applications with scalability and fault tolerance.
- Implemented a modular architecture with Go interfaces, enabling customizable scheduling algorithms, including a load-balancing PVM (Parallel Virtual Machine) layer for optimized resource allocation.
- Designed and implemented REST APIs for both controller and shared nodes, for interaction with the orchestration system via internal and HTTP clients.
- Migration from an in-memory datastore to BoltDB for persistent data storage & reliable system recovery after restart.

Event-Driven Document Collaboration Platform — *Kafka, Node.js, Socket.io, Docker*  [GitHub](#),  [Live](#)

- Built a real-time document collaboration system using an event-driven architecture powered by Apache Kafka.
- Implemented smart deltas with Operational Transformation (OT) to enable conflict-free, low-latency collaborative editing.
- Designed an independently scalable WebSocket & I/O layer, allowing horizontal scaling of live collaboration sessions.
- Developed a modular backend in Node.js and an interactive client in React for seamless multi-user editing.
- Ensured reliable event propagation, ordering, and replay through Kafka topics for fault-tolerant state synchronization.

S3 Based Distributed Cloud Storage Platform — *Go, React.js, Docker, k8s, CI/CD*  [GitHub](#),  [Live](#)

- Built a scalable, S3-compatible cloud storage platform with a React.js frontend and Go-based backend, containerized using Docker and deployed via Kubernetes with automated CI/CD pipelines.
- Designed a vendor-agnostic storage architecture supporting Amazon S3 APIs, MinIO, and Cloudflare R2, enabling flexible multi-provider object storage integration.
- Benchmarked the system against Google Drive, demonstrating 50–70% lower storage cost through optimized object storage utilization and infrastructure design.

ACHIEVEMENTS & CERTIFICATIONS

- AWS Certified Solutions Architect – Associate** — Score: 871/1000
- Designed, deployed, and maintained** self-hosted Kubernetes infrastructure for 2+ years.
- Solved 200+ DSA problems** across LeetCode and CodeChef.
- Top 5 finalist** in hackathons including SIH 2024 and Codecrax 2025.